

Ridge Regression from a Bayesian Viewpoint

STAT 429: Introduction to Time Series
Week 6

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1 Bayesian Ridge Regression

In this lecture, we discuss the Bayesian analogue of ridge regularization. Let us start by recapping ridge regression from the previous lecture.

1.1 Recap: Ridge Regression

Our model from the last two lectures is given by:

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2) + \cdots + \beta_{n-1}(t-(n-1)) + \epsilon_t \quad (1)$$

where, as always, $\epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$. Here $(t-c) = (t-c)_+$ equals 0 if $t \leq c$ and equals $(t-c)$ if $t > c$.

The unknown parameters in this model are $\beta_0, \beta_1, \dots, \beta_{n-1}$ as well as σ .

Alternatively, (12) can be written as:

$$y = X\beta + \epsilon$$

where

$$X = \begin{pmatrix} 1 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 2 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & n-1 & n-2 & \cdot & \cdot & \cdot & 1 \end{pmatrix} \text{ and } \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \cdot \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix}. \quad (2)$$

The ridge regression estimator for β is given by the minimizer of:

$$\sum_{t=1}^n (y_t - \beta_0 - \beta_1(t-1) - \beta_2(t-2) - \cdots - \beta_{n-1}(t-(n-1)))^2 + \lambda (\beta_2^2 + \beta_3^2 + \cdots + \beta_{n-1}^2). \quad (3)$$

The objective function above can also be written as

$$\|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2.$$

It turns out that can be written in closed form using matrix notation. To see this, note first that the gradient of the above objective function with respect to β is given by

$$\nabla \left(\|y - X\beta\|^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2 \right) = -2X^T y + 2X^T X \beta + 2\lambda \begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \cdot \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix}$$

Let J denote the $n \times n$ diagonal matrix whose diagonal entries are $0, 0, 1, \dots, 1$. In other words, the first two diagonal entries of J are 0 and the rest of the diagonal entries equal 1:

$$J = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 1 \end{pmatrix}$$

With this matrix, we can write

$$\nabla \left(\|y - X\beta\|^2 + \lambda \sum_{t=2}^{n-1} \beta_t^2 \right) = -2X^T y + 2X^T X \beta + 2\lambda \begin{pmatrix} 0 \\ 0 \\ \beta_2 \\ \cdot \\ \cdot \\ \cdot \\ \beta_{n-1} \end{pmatrix} = -2X^T y + 2X^T X \beta + 2\lambda J \beta.$$

Setting this gradient equal to zero, we get

$$-2X^T y + 2X^T X \beta + 2\lambda J \beta = 0 \implies (X^T X + \lambda J) \beta = X^T y.$$

which gives

$$\beta = (X^T X + \lambda J)^{-1} X^T y. \quad (4)$$

This looks very similar to the usual linear regression least squares formula $(X^T X)^{-1} X^T y$ with the only difference being the presence of the λJ term.

1.2 Bayesian Regularization

We now treat regularization in high-dimensional regression from the Bayesian point of view. Before discussing regularization, let us first recap the basics of Bayesian regression in the model:

$$y = X\beta + \epsilon \quad \text{with } \epsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

The basic prior that we used previously is

$$\beta_j \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(-C, C).$$

for a large positive constant C . For this prior, we showed (see e.g., problem 5 in Homework 1) that

$$\beta \mid \text{data}, \sigma \sim N \left((X^T X)^{-1} X^T y, \sigma^2 (X^T X)^{-1} \right) \quad (5)$$

when $C \rightarrow \infty$. This fact is not quite true if C is not very large.

A slightly different prior which allows exact formulae even for finite C is the Gaussian prior:

$$\beta_j \stackrel{\text{i.i.d.}}{\sim} N(0, C). \quad (6)$$

Under this prior, it turns out that

$$\beta \mid \text{data}, \sigma \sim N \left(\left(\frac{X^T X}{\sigma^2} + \frac{I}{C} \right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + \frac{I}{C} \right)^{-1} \right) \quad (7)$$

where I is the identity matrix. It is instructive to compare (5) and (7). Unlike (5) which is only true for large C , the fact (7) is true for every $C > 0$. It is also clear that when $C \rightarrow \infty$, then (7) is the same as (5). Observe that when C is large, there is not much difference qualitatively between $\text{unif}(-C, C)$ and $N(0, C)$ (they are both uninformative priors).

We shall prove a more general form of (7) later in this lecture.

Now let us specialize to the case of the high dimensional regression (2). If we use the prior (6) with $C \rightarrow \infty$, then the posterior mean becomes the unregularized least squares (or unregularized MLE) estimator $(X^T X)^{-1} X^T y$. The fitted values will then perfectly interpolate the data leading to overfitting. From the Bayesian perspective, this is happening because the prior (6) with very large C is not useful for this dataset. The prior needs to be changed for a more meaningful analysis. In the frequentist analysis, the main motivation for the ridge regularization (3) is the need to obtain smaller estimates for $\beta_2, \dots, \beta_{n-1}$ which will lead to a smoother fit to the data. This same effect can be obtained by the following modification of the prior (6):

$$\beta_0, \beta_1 \stackrel{\text{i.i.d.}}{\sim} N(0, C) \quad \text{and} \quad \beta_2, \dots, \beta_{n-1} \stackrel{\text{i.i.d.}}{\sim} N(0, \tau^2) \quad (8)$$

for a small parameter τ (in the above, we also assume that $\beta_0, \dots, \beta_{n-1}$ are all independent). The prior (8) can be written as

$$\beta \sim N(0, Q) \quad (9)$$

where Q is the diagonal matrix with diagonal entries $C, C, \tau^2, \dots, \tau^2$. Under the prior (9), the posterior of β is given by

$$\beta \mid \text{data}, \sigma \sim N \left(\left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \right) \quad (10)$$

We will prove this result later in this lecture. The posterior mean therefore is given by

$$\left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2} = (X^T X + \sigma^2 Q^{-1})^{-1} X^T y. \quad (11)$$

This expression is closely related to the ridge estimator (4). Note that Q^{-1} is diagonal with diagonal entries $1/C, 1/C, 1/\tau^2, \dots, 1/\tau^2$. When C is very large, the first two diagonal entries of Q^{-1} are very close to zero so that

$$Q^{-1} \approx \frac{1}{\tau^2} J.$$

Thus the posterior mean (11) is therefore

$$\left(X^T X + \frac{\sigma^2}{\tau^2} J \right)^{-1} X^T y$$

which matches (4) if

$$\lambda = \frac{\sigma^2}{\tau^2} \quad \text{or, equivalently} \quad \tau = \frac{\sigma}{\sqrt{\lambda}}.$$

Ridge regularization therefore can be understood as Bayesian regression with the prior (8). The precise equivalence is obtained if λ is related to τ^2 via $\lambda = \sigma^2/\tau^2$.

1.3 Bayesian approach for dealing with unknown τ and σ

One gets smooth fits to the data by working with the prior (8) for small τ . This is not very surprising because the prior injects a strong amount of bias in favor of smooth fits. The real power of the Bayesian approach lies in the ability to automatically infer τ from the data. This is done by simply placing a prior on τ (along with the priors on β and σ). We shall use the following prior:

$$\log \tau, \log \sigma \stackrel{\text{i.i.d}}{\sim} \text{unif}(-C, C)$$

and

$$\beta \mid \tau, \sigma \sim N(0, Q)$$

where Q is the same as in (9). Note that this prior implies that we are allowing essentially (because C is large) all possible values of τ and σ . In particular, we are **not** *a priori* ruling out large τ just because we don't like wiggly fits.

The prior joint density for β, τ, σ is

$$\begin{aligned} f_{\beta, \tau, \sigma}(\beta, \tau, \sigma) &= f_{\tau}(\tau) f_{\sigma}(\sigma) f_{\beta \mid \tau}(\beta) \\ &= \frac{I\{e^{-C} < \tau < e^C\}}{2C\tau} \frac{I\{e^{-C} < \sigma < e^C\}}{2C\sigma} \left(\frac{1}{\sqrt{2\pi}} \right)^n \frac{1}{\sqrt{\det Q}} \exp \left(-\frac{1}{2} \beta^T Q^{-1} \beta \right) \\ &\propto \frac{I\{e^{-C} < \tau, \sigma < e^C\}}{\tau \sigma} \frac{1}{\sqrt{\det Q}} \exp \left(-\frac{1}{2} \beta^T Q^{-1} \beta \right). \end{aligned}$$

We will also ignore the indicator because C will be very large. It is important to note that Q is not a constant matrix as it depends on τ . The likelihood is (as usual in linear regression)

$$\left(\frac{1}{\sqrt{2\pi}} \right)^n \sigma^{-n} \exp \left(-\frac{1}{2\sigma^2} \|y - X\beta\|^2 \right).$$

The posterior for β, τ, σ is therefore

$$f_{\beta, \tau, \sigma \mid \text{data}}(\beta, \tau, \sigma) \propto \frac{\sigma^{-n-1} \tau^{-1}}{\sqrt{\det Q}} \exp \left(-\frac{1}{2} \left(\frac{1}{\sigma^2} \|y - X\beta\|^2 + \beta^T Q^{-1} \beta \right) \right).$$

The term inside the exponent is a quadratic in β and it is natural to complete the square which is done as follows:

$$\begin{aligned} \frac{1}{\sigma^2} \|y - X\beta\|^2 + \beta^T Q^{-1} \beta &= \frac{y^T y}{\sigma^2} - \frac{2\beta^T X^T y}{\sigma^2} + \beta^T \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right) \beta \\ &= (\beta - \mu)^T \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right) (\beta - \mu) + \frac{y^T y}{\sigma^2} - \mu^T \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right) \mu \end{aligned}$$

where

$$\mu := \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}$$

We thus have

$$\begin{aligned} &\frac{1}{\sigma^2} \|y - X\beta\|^2 + \beta^T Q^{-1} \beta \\ &= (\beta - \mu)^T \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right) (\beta - \mu) + \frac{y^T y}{\sigma^2} - \frac{y^T X}{\sigma^2} \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}. \end{aligned}$$

Plugging this in the posterior formula, we deduce

$$\begin{aligned}
& f_{\beta, \tau, \sigma | \text{data}}(\beta, \tau, \sigma) \\
& \propto \frac{\sigma^{-n-1} \tau^{-1}}{\sqrt{\det Q}} \exp \left(-\frac{1}{2} \left((\beta - \mu)^T \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right) (\beta - \mu) + \frac{y^T y}{\sigma^2} - \frac{y^T X}{\sigma^2} \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2} \right) \right) \\
& = \frac{\sigma^{-n-1} \tau^{-1}}{\sqrt{\det Q}} \exp \left(-\frac{1}{2} (\beta - \mu)^T \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right) (\beta - \mu) \right) \exp \left(-\frac{y^T y}{2\sigma^2} \right) \\
& \times \exp \left(\frac{y^T X}{2\sigma^2} \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2} \right).
\end{aligned}$$

This expression may look complicated but the dependence on β is simple through the quadratic which implies that

$$\beta | \text{data}, \sigma, \tau \sim N \left(\mu, \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \right) = N \left(\left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \right)$$

This proves (7) and (17). It is also straightforward to integrate β from the joint posterior to obtain the posterior of τ, σ :

$$\begin{aligned}
& f_{\tau, \sigma | \text{data}}(\tau, \sigma) \\
& \propto \frac{\sigma^{-n-1} \tau^{-1}}{\sqrt{\det Q}} \sqrt{\det \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1}} \exp \left(-\frac{y^T y}{2\sigma^2} \right) \exp \left(\frac{y^T X}{2\sigma^2} \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2} \right).
\end{aligned}$$

In practice, inference can be carried out by first taking a grid of σ and τ values and computing the above posterior (on the logarithmic scale) at the grid points. We can obtain point estimates of σ and τ by taking the posterior maximizers. Alternatively, we can obtain posterior samples of σ and τ by sampling from the grid points with posterior weights. For each (σ, τ) sample, one can sample β using the multivariate normal distribution (17).

This grid approach can be avoided by using MCMC methods such as the Gibbs sampler. We shall not be discussing these.

1.4 Comments on Bayesian Regularization

In practice, the posterior $f_{\tau, \sigma | \text{data}}(\tau, \sigma)$ tends to prefer τ values which are neither too small nor too large. Because

$$f_{\tau, \sigma | \text{data}}(\tau, \sigma) \propto f_{\text{data} | \tau, \sigma}(\tau, \sigma) f_{\tau, \sigma}(\tau, \sigma),$$

and the prior $f_{\tau, \sigma}(\tau, \sigma)$ is quite flat, the likelihood $f_{\text{data} | \tau, \sigma}(\tau, \sigma)$ must prefer values of τ which are neither too small nor too large. Note that there is a big difference between the two likelihoods:

$$f_{\text{data} | \beta, \sigma}(\text{data}) \quad \text{and} \quad f_{\text{data} | \tau, \sigma}(\text{data}).$$

Maximizing $f_{\text{data} | \beta, \sigma}(\text{data})$ leads to the unregularized least squares estimate which leads to overfitting. On the other hand, maximizing $f_{\text{data} | \tau, \sigma}(\text{data})$ often leads to a fairly small estimate of $\hat{\tau}$ leading to a smooth trend function. The reason for this discrepancy can be understood by noting that

$$f_{\text{data} | \tau, \sigma}(\text{data}) = \int f_{\text{data} | \beta, \sigma}(\text{data}) f_{\beta | \tau}(\beta) d\beta.$$

When τ is large, the term $f_{\beta | \tau}(\beta)$ will be small simply because the normal density with variance τ^2 will be flat for large τ . On the other hand, when τ is too small, the weight $f_{\beta | \tau}(\beta)$ will be significant only for very smooth β s but these β s will have poor values for $f_{\text{data} | \beta, \sigma}(\text{data})$.

1.5 More on Bayesian Regularization

1.5.1 Recap

In the last lecture, we discussed Bayesian regularization in the context of the following high-dimensional linear regression model:

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2) + \cdots + \beta_{n-1}(t-(n-1)) + \epsilon_t \quad (12)$$

The unknown parameters in this model are $\beta_0, \dots, \beta_{n-1}$ as well as σ . We can write this model in regression form as $y = X\beta + \epsilon$ for the very special matrix X :

$$X = \begin{pmatrix} 1 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 1 & 2 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 1 & n-1 & n-2 & \cdot & \cdot & \cdot & 1 \end{pmatrix}$$

We discussed Bayesian inference with the prior (throughout C denotes a very large constant):

$$\beta_0, \beta_1 \stackrel{\text{i.i.d.}}{\sim} \text{unif}(-C, C) \quad \text{and} \quad \beta_2, \dots, \beta_{n-1} \stackrel{\text{i.i.d.}}{\sim} N(0, \tau^2). \quad (13)$$

This prior depends on the unknown parameter (sometimes called 'hyperparameter') τ . We treat τ also as an unknown parameter (along with σ) and place the following prior on τ, σ :

$$\tau, \sigma \stackrel{\text{i.i.d.}}{\sim} \text{unif}(-C, C).$$

The prior assumption (13) can be written in matrix notation as:

$$\beta \mid \tau, \sigma \sim N(0, Q)$$

where Q is the diagonal matrix with diagonal entries $C, C, \tau^2, \dots, \tau^2$.

We calculated the posterior distribution of all the parameters β, σ, τ in the last lecture. This posterior can be described as follows. First the conditional posterior distribution of β given the other two parameters τ, σ is given by:

$$\beta \mid \text{data}, \sigma, \tau \sim N \left(\left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \right) \quad (14)$$

Because Q is diagonal with diagonal entries $C, C, 1/\tau^2, \dots, 1/\tau^2$, it is easy to see that Q^{-1} is also diagonal with diagonal entries $1/C, 1/C, \tau^{-2}, \dots, \tau^{-2}$. Because C is large, we can approximate Q^{-1} by the matrix with diagonal entries $0, 0, \tau^{-2}, \dots, \tau^{-2}$. In other words:

$$Q^{-1} \approx \frac{1}{\tau^2} J \quad \text{where } J = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 1 \end{pmatrix}$$

With this approximation, the conditional posterior of β (given τ, σ) in (14) becomes:

$$\begin{aligned}\beta \mid \text{data}, \sigma, \tau &\sim N \left(\left(\frac{X^T X}{\sigma^2} + \frac{J}{\tau^2} \right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + \frac{J}{\tau^2} \right)^{-1} \right) \\ &\sim N \left(\left(X^T X + \frac{\sigma^2}{\tau^2} J \right)^{-1} X^T y, \sigma^2 \left(X^T X + \frac{\sigma^2}{\tau^2} J \right)^{-1} \right)\end{aligned}\quad (15)$$

We saw in the last lecture that the mean of this conditional posterior distribution:

$$\left(X^T X + \frac{\sigma^2}{\tau^2} J \right)^{-1} X^T y$$

coincides with the Ridge Regularized estimate with tuning parameter λ provided $\lambda = \sigma^2/\tau^2$.

We also derived that the posterior of τ and σ is given by the formula:

$$\begin{aligned}f_{\tau, \sigma \mid \text{data}}(\tau, \sigma) \\ \propto \frac{\sigma^{-n-1} \tau^{-1}}{\sqrt{\det Q}} \sqrt{\det \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1}} \exp \left(-\frac{y^T y}{2\sigma^2} \right) \exp \left(\frac{y^T X}{2\sigma^2} \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2} \right).\end{aligned}$$

Using again the approximation $Q^{-1} \approx (1/\tau^2)J$ as well as $\det Q = C^2 \tau^{2(n-2)} \propto \tau^{2(n-2)}$, we get

$$\begin{aligned}f_{\tau, \sigma \mid \text{data}}(\tau, \sigma) \\ \propto \sigma^{-n-1} \tau^{-n+1} \sqrt{\det \left(\frac{X^T X}{\sigma^2} + \frac{J}{\tau^2} \right)^{-1}} \exp \left(-\frac{y^T y}{2\sigma^2} \right) \exp \left(\frac{y^T X}{2\sigma^2} \left(\frac{X^T X}{\sigma^2} + \frac{J}{\tau^2} \right)^{-1} \frac{X^T y}{\sigma^2} \right) \\ = \sigma^{-1} \tau^{-n+1} \sqrt{\det \left(X^T X + \frac{\sigma^2}{\tau^2} J \right)^{-1}} \exp \left(-\frac{y^T y}{2\sigma^2} \right) \exp \left(\frac{y^T X}{2\sigma^2} \left(X^T X + \frac{\sigma^2}{\tau^2} J \right)^{-1} X^T y \right)\end{aligned}$$

This posterior lets us figure out the value of τ (as well as σ) from the data. Specifically, we can take a grid of σ and τ values and compute the above posterior (on the logarithmic scale) at the grid points. We then obtain point estimates of σ and τ by taking the posterior mode or mean. Alternatively, we can obtain posterior samples of σ and τ by sampling from the grid points with posterior weights. For each (σ, τ) sample, one can sample β using the multivariate normal distribution (17).

In most cases, the posterior $f_{\tau, \sigma \mid \text{data}}(\tau, \sigma)$ will prefer values of τ that are not too large. Note that large τ would mean that the resulting β estimates (see (15)) will lead to fitted values that overfit the data. In other words, this Bayesian approach for selecting τ automatically avoids overfitting.

Further, unless the data is very simple (in the sense of being well-explained by a single line), the posterior $f_{\tau, \sigma \mid \text{data}}(\tau, \sigma)$ will also prefer values of τ that are not too small. Note that small τ will mean that the resulting fitted values will be very close to the least squares line.

Thus, this Bayesian approach offers protection from both overfitting and underfitting. To get some insight as to why $f_{\tau, \sigma \mid \text{data}}(\tau, \sigma)$ protects from overfitting and underfitting, consider the following alternative expression for it:

$$f_{\tau, \sigma \mid \text{data}}(\tau, \sigma) \propto f_{\text{data} \mid \tau, \sigma}() f_{\tau, \sigma}(\tau, \sigma).$$

In the above, $f_{\tau, \sigma}(\tau, \sigma)$ is a very flat function (because we are using uninformative priors for τ and σ) so the behavior of $f_{\tau, \sigma \mid \text{data}}(\tau, \sigma)$ will be mainly driven by the first term: $f_{\text{data} \mid \tau, \sigma}()$.

This term can be seen as the likelihood of the data only in terms of τ and σ . It is related to the original likelihood $f_{|\beta,\sigma}()$ via

$$f_{|\tau,\sigma}() = \int f_{|\beta,\sigma}() f_{\beta|\tau}(\beta) d\beta. \quad (16)$$

This integrated likelihood $f_{|\tau,\sigma}()$ tends to prefer values of τ which are neither too small nor too large. To understand why, recall that

$$f_{|\tau,\sigma}() = \int f_{|\beta,\sigma}() f_{\beta|\tau}(\beta) d\beta.$$

Thus, for a given value of τ , the integrated likelihood is averaging the data likelihood over values of β that are plausible under the prior corresponding to that τ .

Suppose first that τ is very small. Then the prior $f_{\beta|\tau}(\beta)$ is concentrated on values of β for which $\beta_2, \dots, \beta_{n-1}$ are close to zero, and hence the corresponding fitted values are close to a single line. If the data contain substantial nonlinear structure, most of these fitted curves will fit the data poorly, so $f_{|\beta,\sigma}()$ will be small for most values of β receiving substantial prior probability. Consequently, the integrated likelihood will be small.

Now suppose that τ is very large. In this case, the prior allows a very large collection of possible fitted curves, including very irregular ones. Some particular choices of β may fit the observed data extremely well. However, the integrated likelihood does not choose the best-fitting value of β ; it averages the likelihood over all values of β according to the prior $f_{\beta|\tau}$. When τ is very large, the prior probability is spread over a very large region of the parameter space. Only a relatively small portion of that region will correspond to curves that fit this particular observed dataset well. Much of the prior probability is therefore placed on values of β having relatively small data likelihood, which brings down the average in (16).

Thus, very small values of τ can be disfavored because they restrict the model too strongly and lead to underfitting, while very large values of τ can be disfavored because they spread prior probability over too many possible complicated fits. Intermediate values of τ can put substantial prior probability on fitted curves flexible enough to explain the structure in the data without allowing an unnecessarily large range of possibilities.

It is important to note that the Bayesian approach for selecting hyperparameters such as τ is very different from the frequentist approach for selecting the tuning parameter λ using Cross-Validation (CV). CV relies on splitting the data into two parts: training and test, while the Bayesian calculation uses the entire data (no splits are necessary). For more on the relation between Bayesian approaches for hyperparameter tuning and model selection, and frequentist methods, see http://www.inference.org.uk/mackay/Bayes_FAQ.html#gcv and <https://statmodeling.stat.columbia.edu/2011/12/04/david-mackay-and-occams-razor/>.

1.5.2 Calculations with a slightly different prior

The Bayesian method described above involves placing a grid on both τ and σ . This grid approach can be avoided by using MCMC methods such as the Gibbs sampler. We shall not be discussing these (see the 248 problem in Homework 3 if you are interested in Gibbs sampling).

There is a slightly different prior which lets us integrate over σ in closed form. So the grid approach would then be needed only for τ (and not σ) which reduces the computational burden to some extent. This method is described here.

The idea is to reparametrize τ as $\tau = \sigma\gamma$ for a new parameter γ , and change the prior to

$$\log \gamma, \log \sigma \stackrel{\text{i.i.d}}{\sim} \text{unif}(-C, C).$$

and

$$\beta \mid \gamma, \sigma \sim N(0, Q)$$

where Q is the diagonal matrix with diagonal entries $C, C, \gamma^2 \sigma^2, \dots, \gamma^2 \sigma^2$ (this is the same Q as before; the only difference is that now we are writing $\gamma\sigma$ for τ). The prior joint density for β, γ, σ now becomes:

$$\begin{aligned} f_{\beta, \gamma, \sigma}(\beta, \gamma, \sigma) &= f_{\gamma}(\gamma) f_{\sigma}(\sigma) f_{\beta \mid \gamma, \sigma}(\beta) \\ &= \frac{I\{e^{-C} < \gamma < e^C\}}{2C\gamma} \frac{I\{e^{-C} < \sigma < e^C\}}{2C\sigma} \left(\frac{1}{\sqrt{2\pi}} \right)^n \frac{1}{\sqrt{\det Q}} \exp \left(-\frac{1}{2} \beta^T Q^{-1} \beta \right) \\ &\propto \frac{I\{e^{-C} < \gamma, \sigma < e^C\}}{\gamma\sigma} \frac{1}{\sqrt{\det Q}} \exp \left(-\frac{1}{2} \beta^T Q^{-1} \beta \right). \end{aligned}$$

This new parameter γ is almost directly related to the tuning parameter λ in ridge regression. Specifically, $\lambda = 1/\gamma^2$ or $\gamma = 1/\sqrt{\lambda}$.

The likelihood function is exactly the same as before:

$$\left(\frac{1}{\sqrt{2\pi}} \right)^n \sigma^{-n} \exp \left(-\frac{1}{2\sigma^2} \|y - X\beta\|^2 \right).$$

So the posterior for β, γ, σ is:

$$f_{\beta, \gamma, \sigma \mid \text{data}}(\beta, \gamma, \sigma) \propto \frac{\sigma^{-n-1} \gamma^{-1}}{\sqrt{\det Q}} \exp \left(-\frac{1}{2} \left(\frac{1}{\sigma^2} \|y - X\beta\|^2 + \beta^T Q^{-1} \beta \right) \right).$$

From here, we proceed exactly as in last lecture to derive:

$$\beta \mid \text{data}, \sigma, \gamma \sim N \left(\left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2}, \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \right) \quad (17)$$

Again, as in last lecture, we integrate β from the joint posterior to obtain the posterior of γ, σ :

$$\begin{aligned} f_{\gamma, \sigma \mid \text{data}}(\gamma, \sigma) &\propto \frac{\sigma^{-n-1} \gamma^{-1}}{\sqrt{\det Q}} \sqrt{\det \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1}} \exp \left(-\frac{y^T y}{2\sigma^2} \right) \exp \left(\frac{y^T X}{2\sigma^2} \left(\frac{X^T X}{\sigma^2} + Q^{-1} \right)^{-1} \frac{X^T y}{\sigma^2} \right). \end{aligned}$$

Simplifying by use of $\det Q \propto (\gamma^2 \sigma^2)^{n-2}$ and $Q^{-1} \approx J/(\gamma^2 \sigma^2)$, we can write

$$\begin{aligned} f_{\gamma, \sigma \mid \text{data}}(\gamma, \sigma) &\propto \sigma^{-2n+1} \gamma^{-n+1} \sqrt{\det \left(\frac{X^T X}{\sigma^2} + \frac{J}{\gamma^2 \sigma^2} \right)^{-1}} \exp \left(-\frac{y^T y}{2\sigma^2} \right) \exp \left(\frac{y^T X}{2\sigma^2} \left(\frac{X^T X}{\sigma^2} + \frac{J}{\gamma^2 \sigma^2} \right)^{-1} \frac{X^T y}{\sigma^2} \right) \\ &\propto \sigma^{-2n+1} \gamma^{-n+1} \sqrt{\sigma^{2n} \det \left(X^T X + \frac{J}{\gamma^2} \right)^{-1}} \exp \left(-\frac{y^T y}{2\sigma^2} \right) \exp \left(\frac{y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y}{2\sigma^2} \right) \\ &= \sigma^{-n+1} \gamma^{-n+1} |X^T X + \gamma^{-2} J|^{-1/2} \exp \left(-\frac{y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y}{2\sigma^2} \right) \end{aligned}$$

The advantage of this parametrization (i.e., in terms of γ, σ as opposed to τ, σ) is that the conditional posterior distribution of σ given γ (and the data) can be written in closed form. This is because from the above expression, we can write

$$f_{\sigma \mid \text{data}, \gamma}(\sigma) \propto \sigma^{-n+1} \exp \left(-\frac{y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y}{2\sigma^2} \right).$$

The right hand side above is actually the pdf of an inverse gamma density (see https://en.wikipedia.org/wiki/Inverse-gamma_distribution). This can be seen by converting it into the density of $1/\sigma^2$:

$$\begin{aligned} f_{1/\sigma^2|\text{data},\gamma}(x) &\propto f_{\sigma|\text{data},\gamma}(x^{-1/2})x^{-3/2} \\ &\propto (x^{-1/2})^{-n+1} \exp\left(-x \frac{y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y}{2}\right) x^{-3/2} \\ &= x^{(n-4)/2} \exp\left(-x \frac{y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y}{2}\right). \end{aligned}$$

Thus

$$\frac{1}{\sigma^2} | \text{data}, \gamma \sim \text{Gamma}\left(\frac{n}{2} - 1, \frac{y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y}{2}\right) \quad (18)$$

Finally, we can marginalize σ to obtain the posterior of γ alone as follows:

$$\begin{aligned} f_{\gamma|\text{data}}(\gamma) &\propto \int_0^\infty \sigma^{-n+1} \gamma^{-n+1} |X^T X + \gamma^{-2} J|^{-1/2} \exp\left(-\frac{y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y}{2\sigma^2}\right) d\sigma \\ &= \gamma^{-n+1} |X^T X + \gamma^{-2} J|^{-1/2} \int_0^\infty \sigma^{-n+1} \exp\left(-\frac{y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y}{2\sigma^2}\right) d\sigma. \end{aligned}$$

Letting

$$A := y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y,$$

we get

$$f_{\gamma|\text{data}}(\gamma) \propto \gamma^{-n+1} |X^T X + \gamma^{-2} J|^{-1/2} \int_0^\infty \sigma^{-n+1} \exp\left(-\frac{A}{2\sigma^2}\right) d\sigma$$

By the change of variable $\sigma = s\sqrt{A}$, we obtain

$$\begin{aligned} f_{\gamma|\text{data}}(\gamma) &\propto \gamma^{-n+1} |X^T X + \gamma^{-2} J|^{-1/2} A^{-(n/2)+1} \int_0^\infty s^{-n+1} \exp\left(-\frac{1}{2s^2}\right) ds \\ &\propto \gamma^{-n+1} |X^T X + \gamma^{-2} J|^{-1/2} A^{-(n/2)+1} \\ &= \frac{\gamma^{-n+1} |X^T X + \gamma^{-2} J|^{-1/2}}{\left(y^T y - y^T X (X^T X + \gamma^{-2} J)^{-1} X^T y\right)^{(n/2)-1}}. \end{aligned}$$

With this, inference can be carried out by first taking a grid of γ values and computing the above posterior (on the logarithmic scale) at the grid points. This posterior can be used to obtain posterior samples of γ . For each sample of γ , we can then sample σ using the distribution (18). Given samples from both γ and σ , we can then sample β using (17).

In the connection with usual ridge regression, we noted previously that $\lambda = \sigma^2/\tau^2$. With the new parametrization $\tau^2 = \gamma^2\sigma^2$, we have $\lambda = 1/\gamma^2$ so that $\gamma = 1/\sqrt{\lambda}$.